COURSE: BASICS OF ELECTRICAL ENGINEERING (EE123ATD)

UNIT 5

Electrical Power Generation



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Electricity

Electricity is a secondary source of energy

- We must use a primary source of energy to transform it into electricity



❖Electricity is the flow of electrical power or charge.

❖It is a **secondary energy source** which means that we get it from the conversion of other sources of energy, like coal, natural gas, oil, nuclear power and other natural sources, which are called primary sources.

❖The energy sources we use to make electricity can be renewable or non-renewable, but **electricity itself is neither renewable nor non-renewable**.

Energy Transformations

Remember what we know about energy transformations:

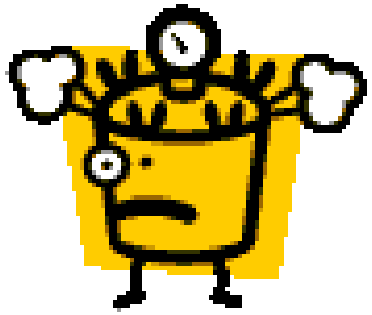
Potential Energy Forms	Kinetic Energy Forms
Chemical Energy	Electrical Energy
Stored Mechanical Energy	Radiant Energy (light and solar)
Nuclear Energy	Thermal Energy
Gravitational Energy	Motion Energy
	Sound Energy

Energy Transformations

Energy Transformations To Create Electricity

Potential Chemical Energy → Kinetic Thermal Energy → Kinetic Motion Energy → Kinetic Electrical Energy







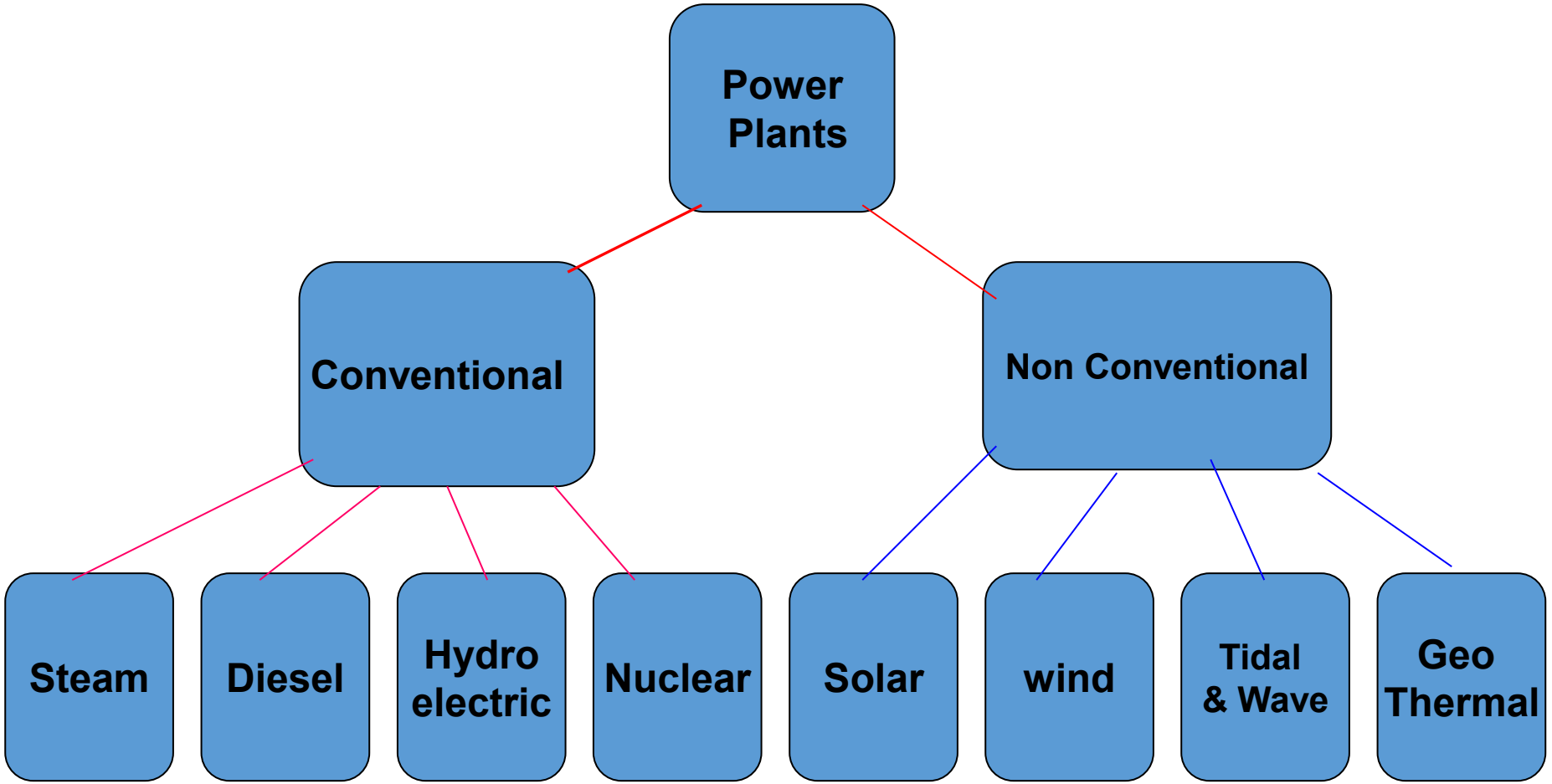


Power Plant

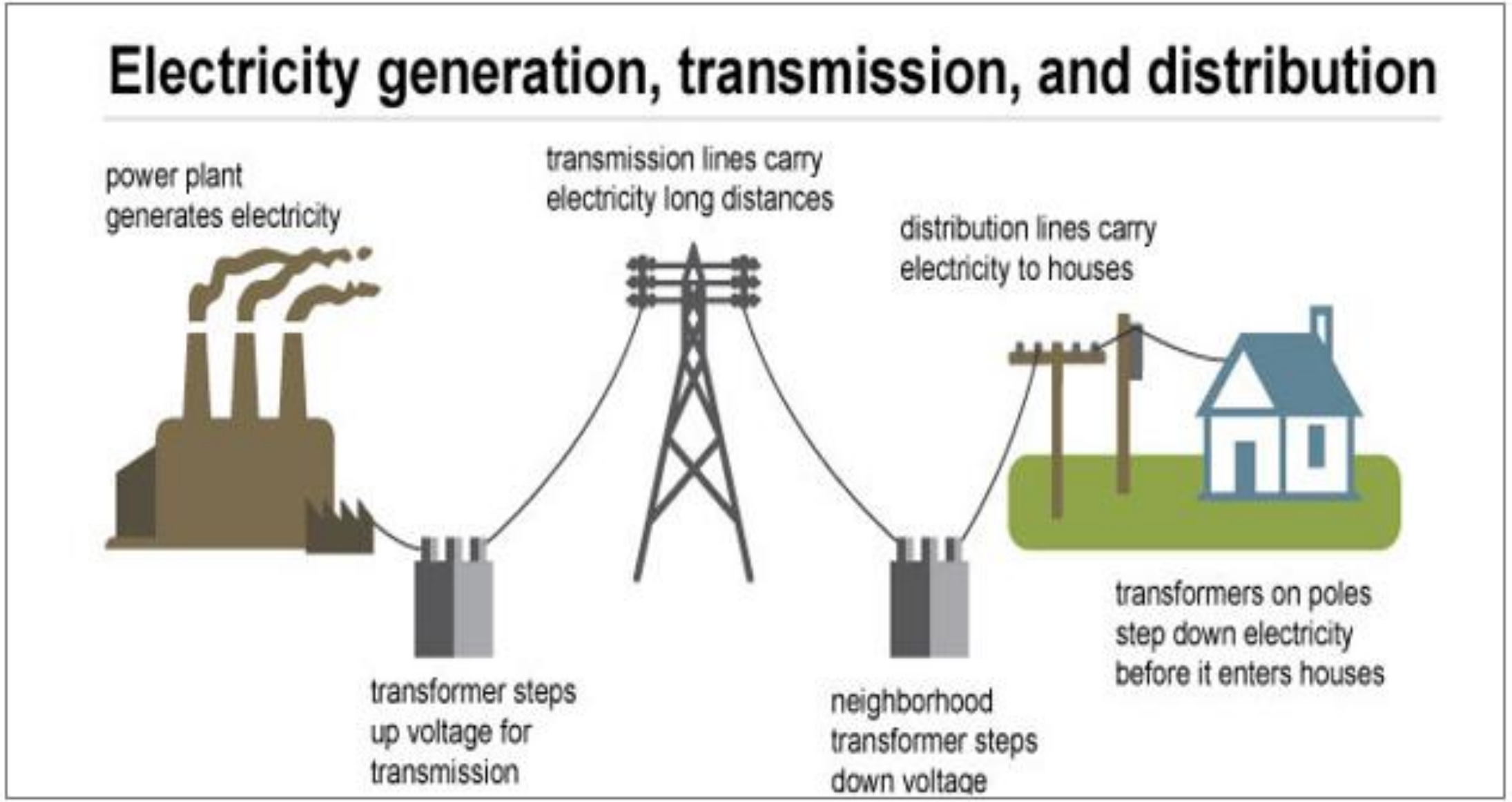
A power station (also referred to as a generating station, power plant, powerhouse or generating plant) is an industrial facility for the generation of electric power.

Hydraulic Energy / Thermal Energy -> Mechanical Energy -> Electrical Energy

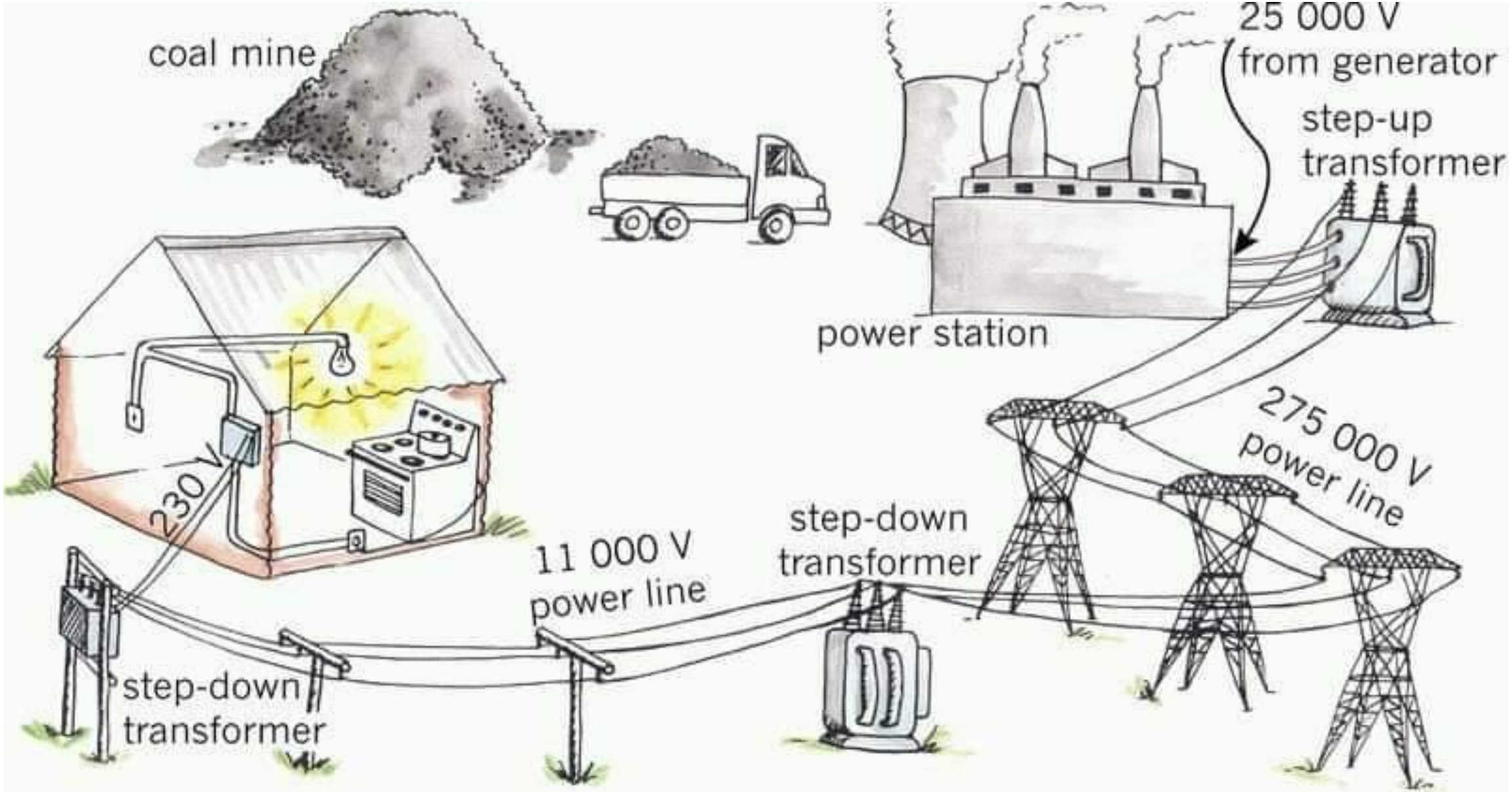
Classification of Power Plants



Electrical Network and its various components



Electrical Network and its various components



How electricity is made and delivered to Bangalore

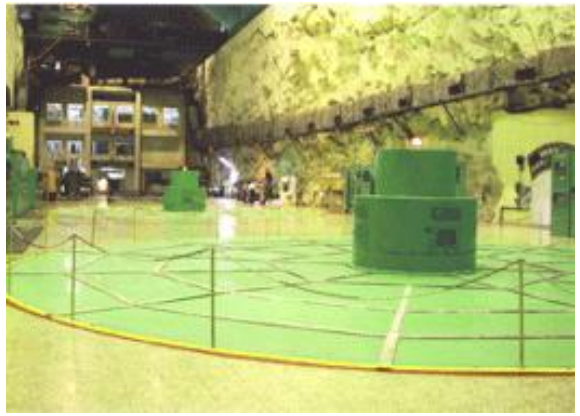
KPCL>>>>> KPTCL>>>>SLDC>>>>>BESCOM





Power Station	Type	Number of Units	Capacity (MW)	Total
Raichur Thermal Power station	Thermal	7 + 1	7x210 + 1x250	1720
Bellary Thermal Power station	Thermal	2 + 1	2x500 + 1x700	1700
Yermarus Thermal Power Station	Thermal	2	2x800	1600
Sharavathi	Hydel	10	103.5	1035
Nagjhari-2	Hydel	5	150	750
Varahi River	Hydel	4	115	460
Almatti Dam	Hydel	5 + 1	5x55 + 1x15	290
Nagjhari-1	Hydel	1	1x150	150
Gerusoppa	Hydel	4	60	240
Kadra Dam	Hydel	3	50	150







How electricity is made and delivered to Bangalore

KPCL>>>>> KPTCL>>>>SLDC>>>>>BESCOM

Bangalore's daily power demand is about 2300 MW (Mega Watt). That is, 2300 MW of power is transmitted throughout the day to the city on average. Bangalore consumes about one-third of the state's total power. Karnataka's average demand is 6000 MW per day.

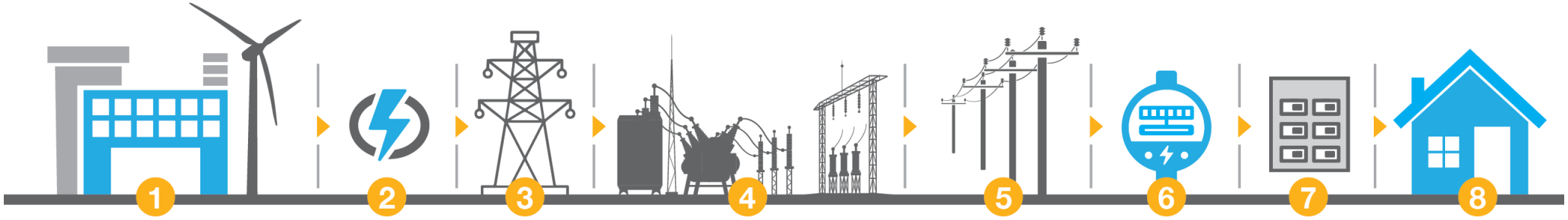
Three agencies are involved in the procurement, transmission and supply of power, before it reaches consumers. Karnataka Power Corporation Ltd (KPCL) is the state agency that gets power from different generating stations. KPCL also buys power from other states when required.

Bangalore is powered by the same grid that supplies to the entire state – there are no specific sources for Bangalore alone.

Another agency, Karnataka Power Transmission Corporation Ltd (KPTCL) is in charge of transmitting power to different ESCOMs, including BESCOM. Once it gets the power from KPTCL, BESCOM's local network supplies it to consumers.

All of this is co-ordinated by KPTCL's State Load Despatch Centre (SLDC). ESCOMs inform SLDC about their power requirement forecast for the next day, 24 hours earlier. Similarly, KPCL informs SLDC of its generation forecast, a day before. Depending on this, the total power is distributed among each ESCOM for the next day. Demand and supply varies through the day, and SLDC maintains real-time data on this.

How electricity is made and delivered to your home



1.Electricity is made at a generating station by huge generators. Generating stations can use wind, coal, natural gas, or water.

2.The current is sent through transformers to increase the voltage to push the power long distances.

3.The electrical charge goes through high-voltage transmission lines that stretch across the country.

4.It reaches a substation, where the voltage is lowered so it can be sent on smaller power lines.

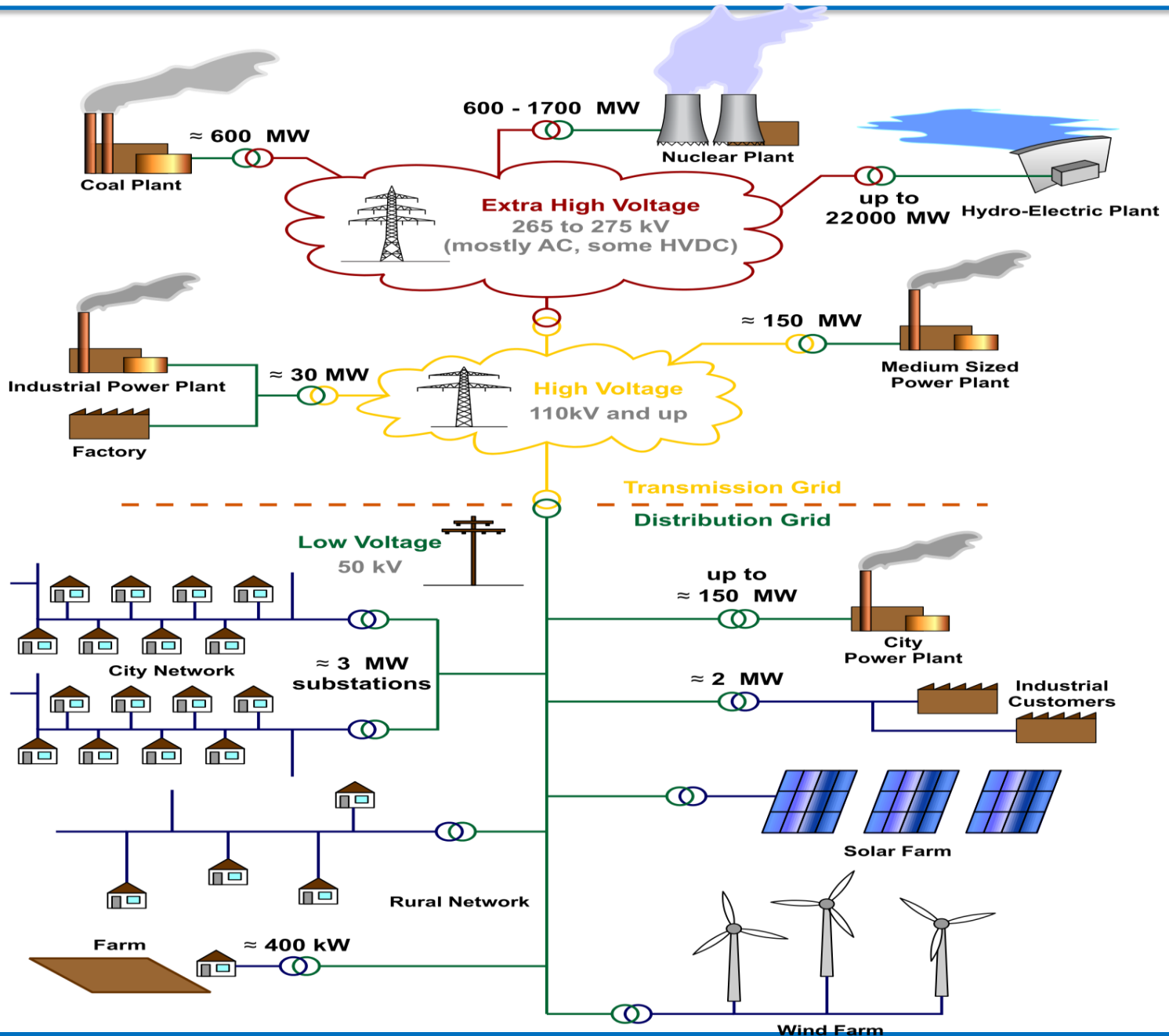
5.It travels through distribution lines to your neighborhood. Smaller transformers reduce the voltage again to make the power safe to use in our homes. These smaller transformers may be mounted on the poles, or sitting on the ground (they're the big green boxes, called pad mount transformers).

6.It connects to your house and passes through a meter that measures how much your family uses.

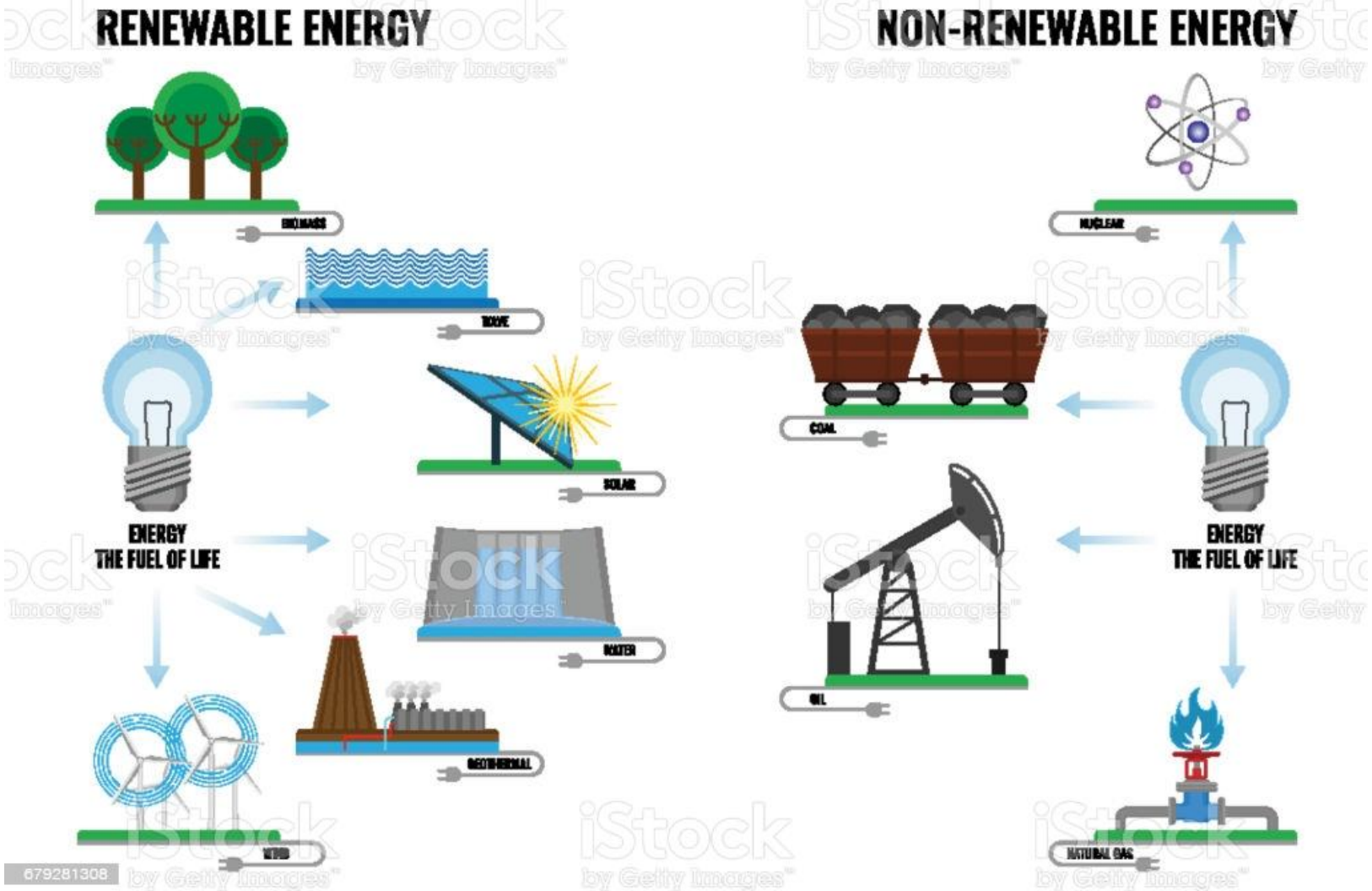
7.The electricity goes to the service panel in your basement or garage, where breakers or fuses protect the wires inside your house from being overloaded. (Never touch a service panel! It is only to be operated by your parents or a professional.)

8.The electricity travels through wires inside the walls to the outlets and switches all over your house.

Electrical Network and its various components

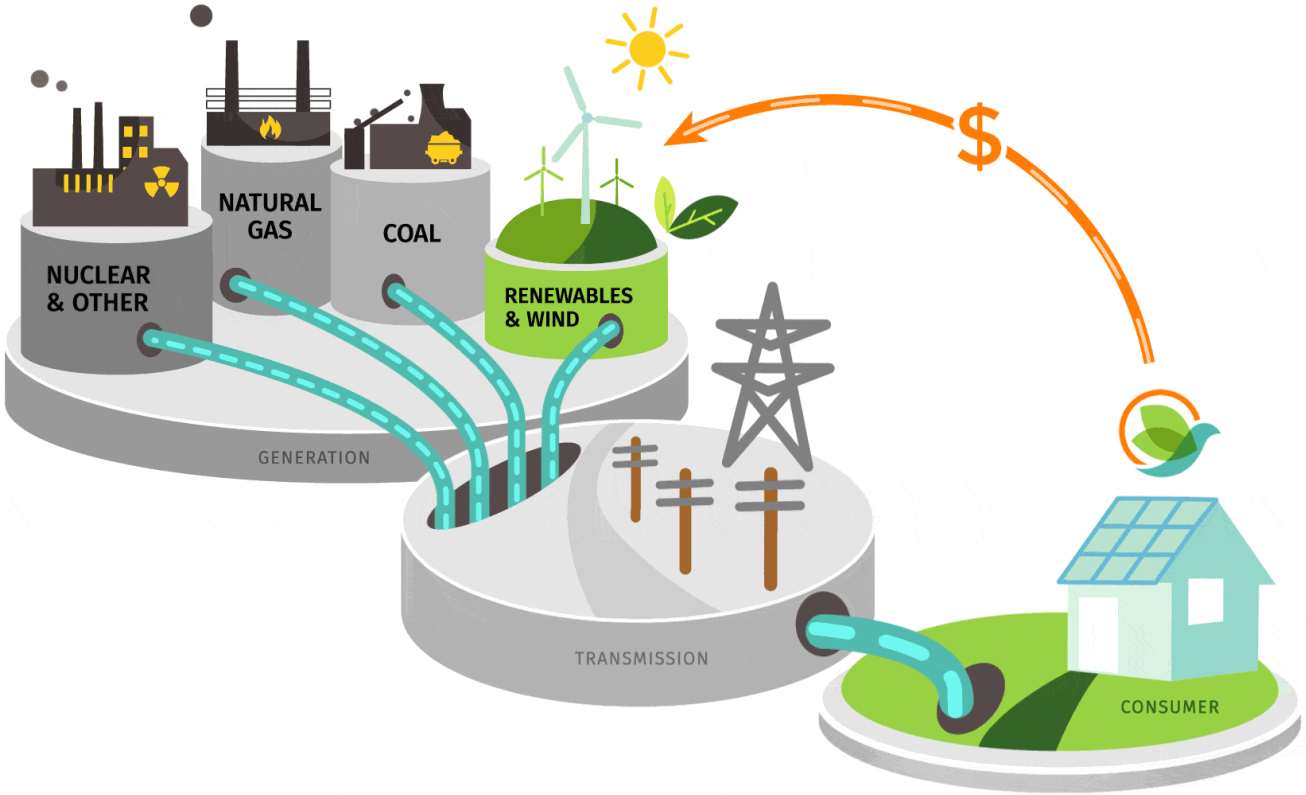


ENERGY SOURCES



Electrical Network and its various components



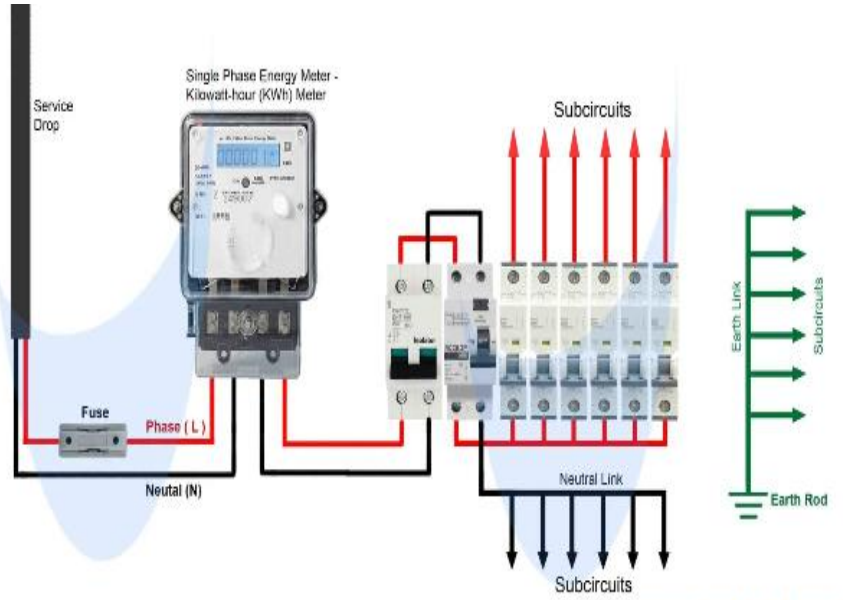


UNIT 5 (PART – 2)

TOPIC:-Fuse and MCB, Electric shock, Earthing, Precautions, Common Household Appliances, Energy Calculations



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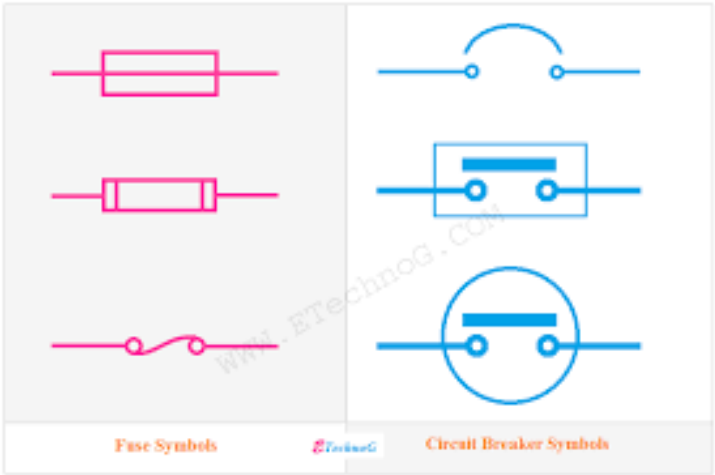
Fuses and MCBs

- The electrical equipment's are designed to carry a particular rated value of current under normal circumstances.
- Under abnormal conditions such as short circuit, overload or any fault the current raises above this value, damaging the equipment and sometimes resulting in fire hazard.
- Overcurrent Protection devices such as Fuses and MCBs are essential in electrical Systems to limit the threats human life and to avoid property Damage.









Fuses: Working, Types, Characteristics, Advantages and Dis advantages

- Fuse is a safety device used in any electrical installation, which forms the weakest link between the supply and the load.
- It is a short length of wire made of lead / tin /alloy of lead and tin/ zinc having a low melting point and low ohmic losses.
- Under normal operating conditions it is designed to carry the full load current.
- If the current increases beyond this designed value due any of the reasons mentioned above, the fuse melts (said to be blown) isolating the power supply from the load .
- Terms Related with Fuses
- Rated current: It is the maximum current, which a fuse can carry without undue heating or melting.
- Fusing current: The minimum current at which the fuse melts is known as the fusing current. It depends on the material characteristics, length, diameter, cross-sectional area of the fuse element and the type of enclosure used.
- Fusing Factor: It is the ratio of the minimum fusing current to the rated current. It is always greater than unity.

The material used for fuse wires must have the following characteristics

Low melting point, Low ohmic losses, High conductivity, Lower rate of deterioration

Advantages of Fuses:

Fast acting, Highly reliable, Relatively cheaper in comparison to other high current interrupting device

Disadvantages of Fuses: Requires replacement, The associated high temperature rise will affect the performance of other devices.

PACKAGES AND MATERIAL

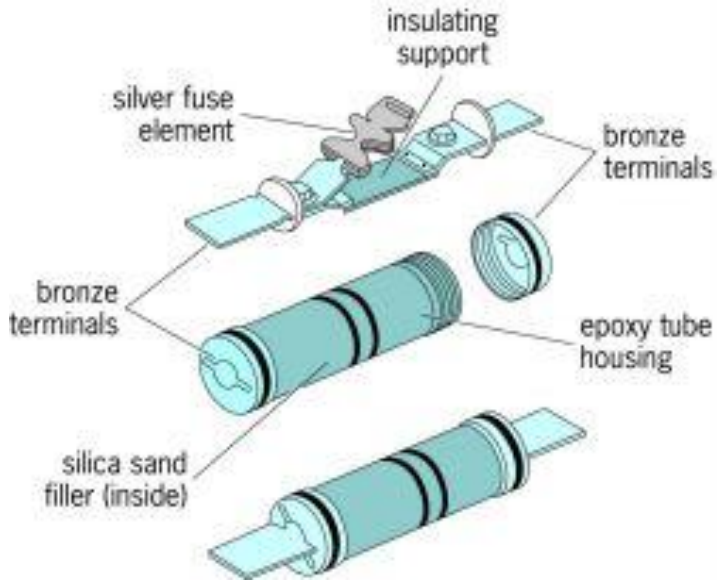
Fuse consist of two main parts:

- Fuse casing
- Fuse elements

Fuse casing may be made up of:

- Ceramics
- Glass
- Plastic
- Molded mica laminates





FUSE ELEMENT MATERIALS

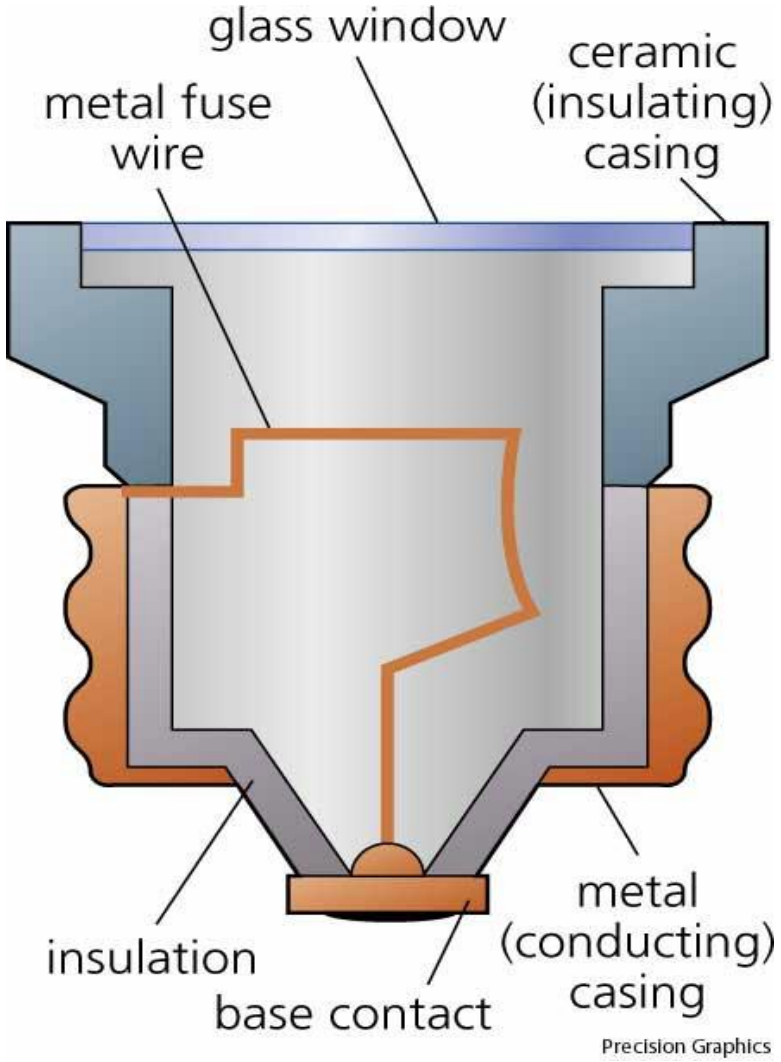
- The materials used for fuse elements must be of low melting point , low ohmic loss ,high conductivity , low cost & free from deterioration.

METAL	MELTING POINT IN °C	SPECIFIC RESISTANE IN μΩ-mm	VALUE OF FUSE CONSTANT K FOR d in m
SILVER	980	16	-
TIN	240	112	12.8
ZINC	419	60	-
LEAD	328	210	10.8
COPPER	1090	17	80
ALUMINIUM	665	28	59

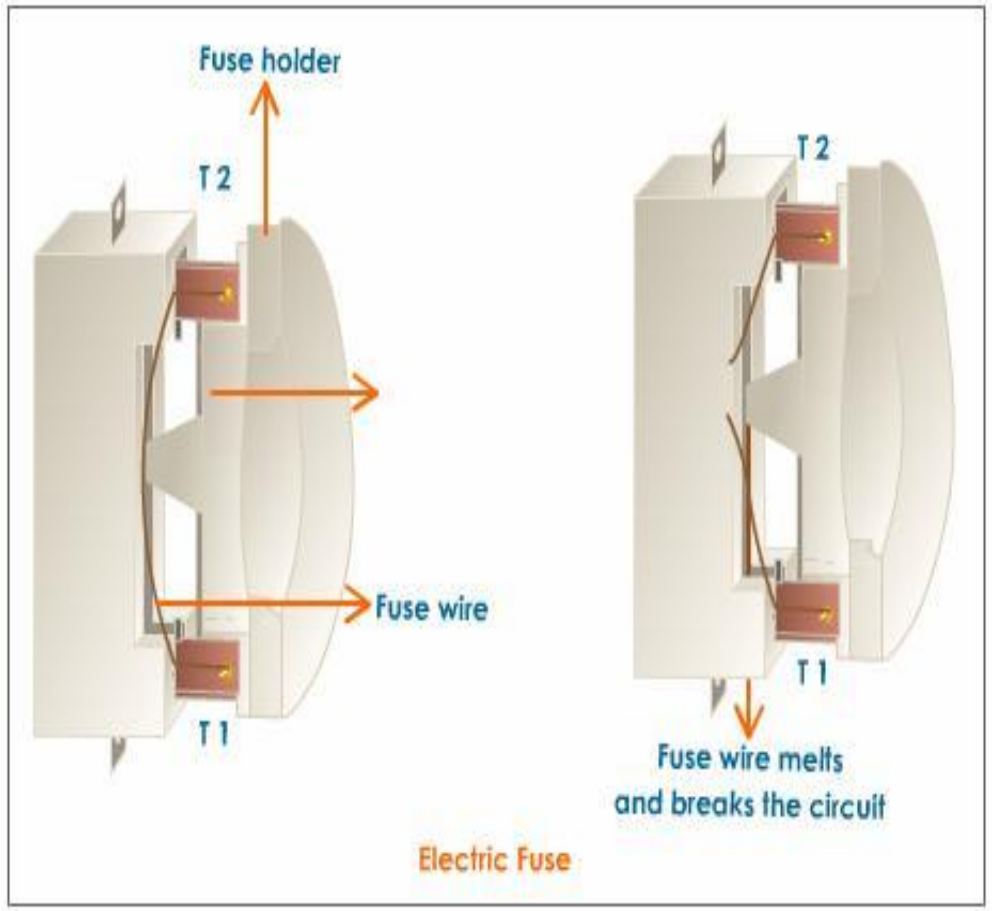
HOW FUSE IS CONSTRUCTED ?

A fuse consists of a metal strip or wire fuse element, of small cross-section compared to the circuit conductors, mounted between a pair of electrical terminals, and enclosed by a non-combustible housing.

The fuse element is made of zinc, copper, silver, aluminum, or alloys to provide stable and predictable characteristics.

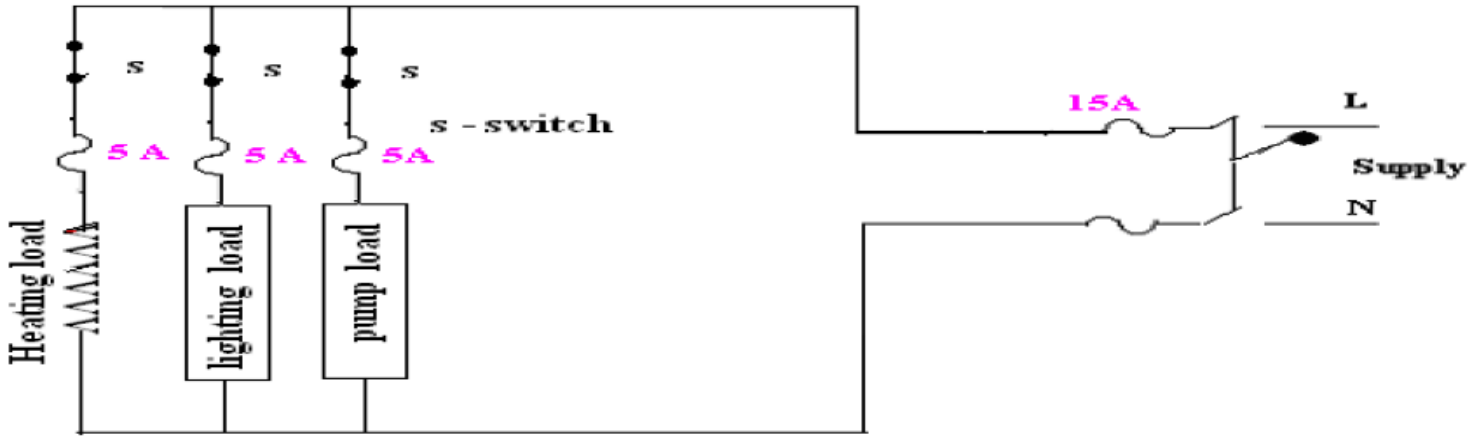


HOW FUSE WORKS ?

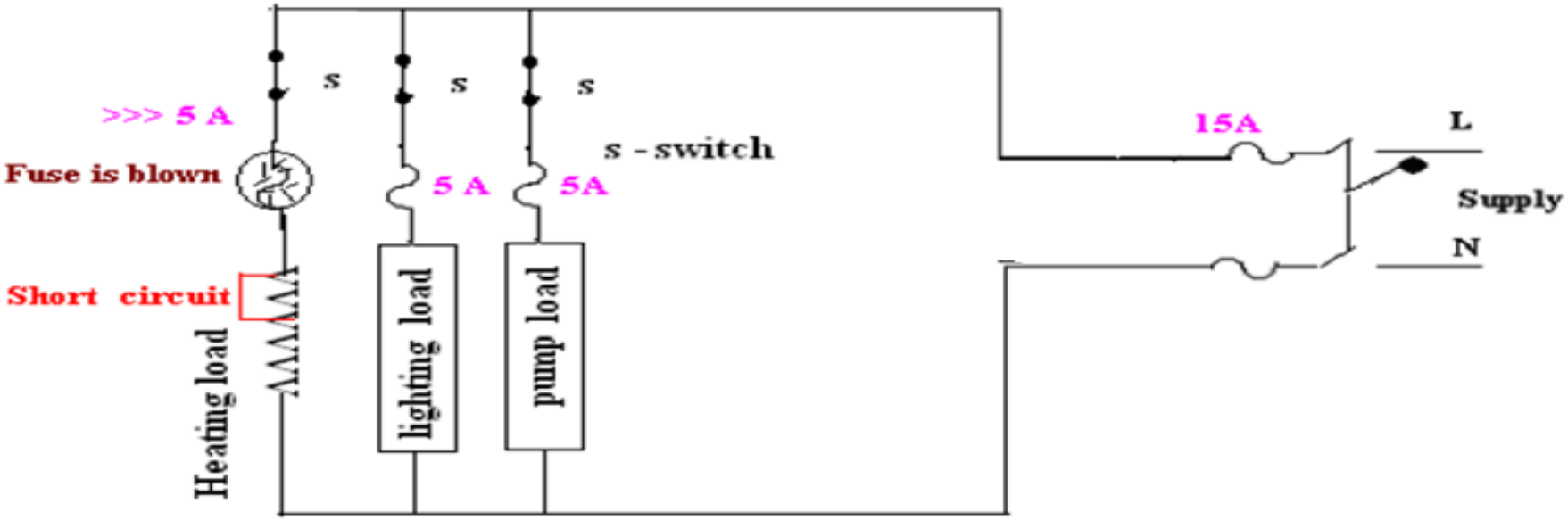


- A fuse has a metallic strip of small cross section area , mounted between a pair of electrical terminal.
- The fuse is connected in series with the circuit through which all the current will pass.
- In case of heavy current, fuse will melt. This will break the circuits and protect the electrical appliance.

HOW FUSE WORKS ?

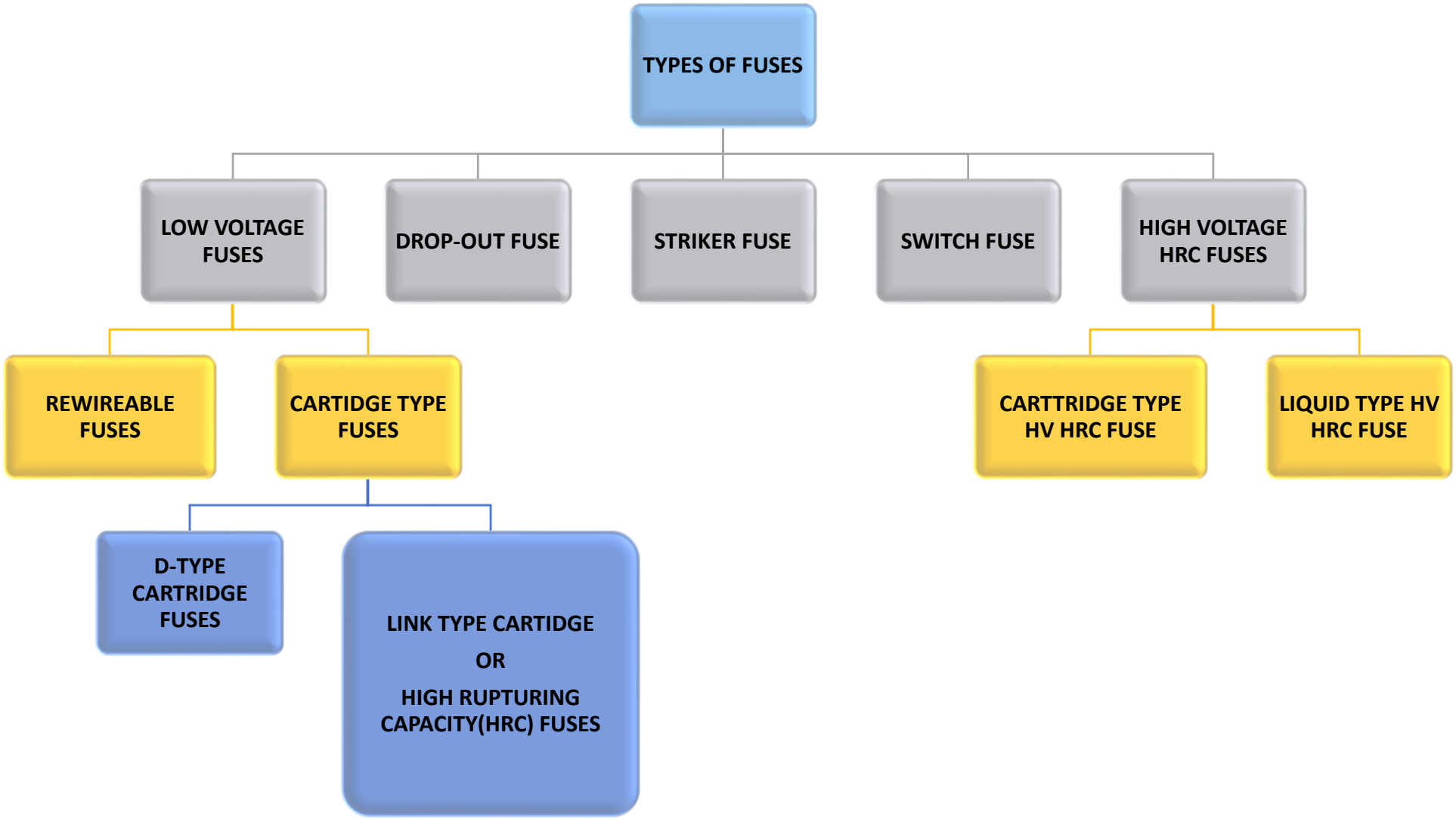


Under normal conditions



Under abnormal conditions

TYPES OF FUSES



LOW VOLTAGE FUSE

- **REWIREABLE FUSES :-**

  The most commonly used fuse in ‘house wiring ’ and a small current circuits is the semi-enclosed or rewirable fuse.

- Also known as KIT-KAT type fuse.

DISADVANTAGES:-

- Unreliable Operation
- Small Time Lag
- Low Rupturing Capacity
- No-Current-Limiting Features
- Slow Speed Of Operation
- Risk Of Flame & Fire



Fuses: Working (video)

ADVANTAGES OF FUSE

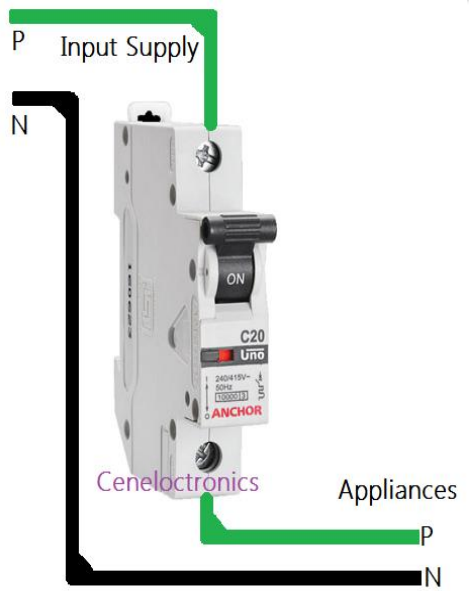
- It is the cheapest form of protection available.
- It needs no maintenance.
- It interrupts enormous short circuit currents without noise , flame , gas or smoke.
- The minimum time of operation can be made much smaller than that with the circuit breakers.
- The smaller size of fuse element imposes a current limiting effect under short circuit.
- Its inverse time-current characteristics enables its use for overload protection.

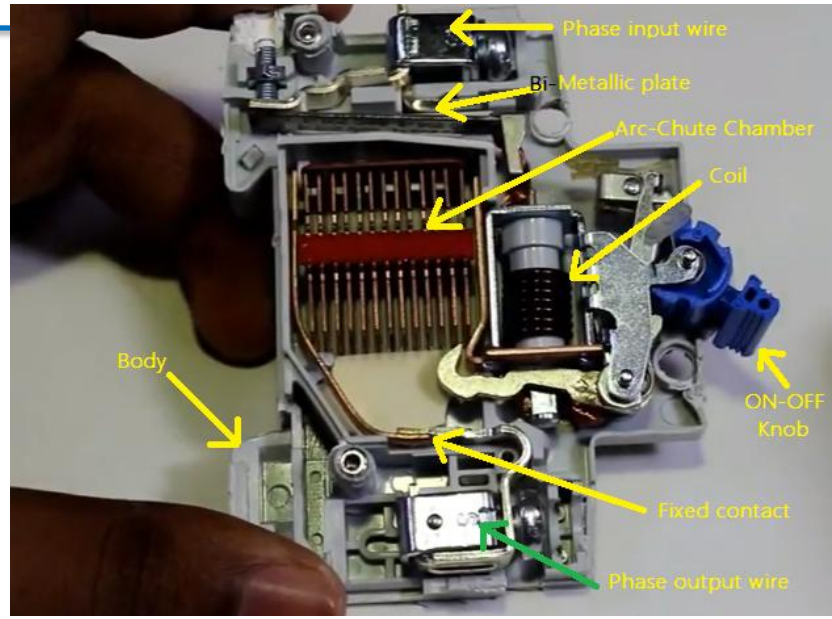
DISADVANTAGES OF FUSE

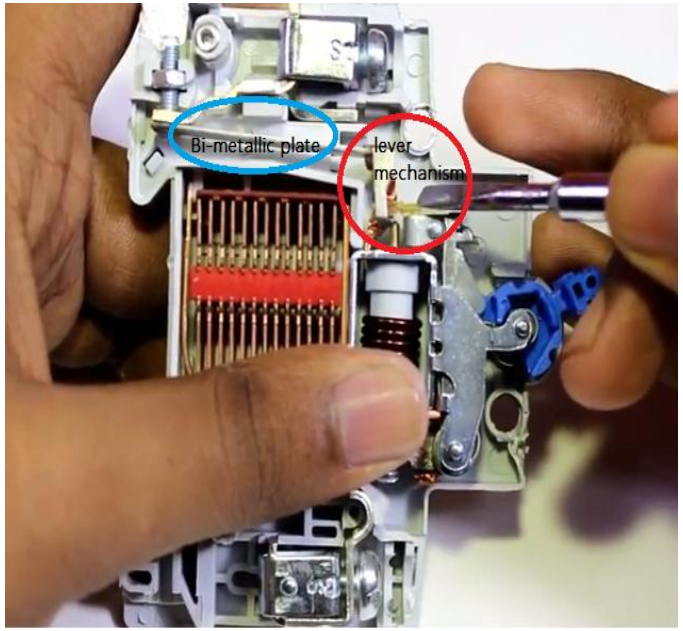
- Considerable time is lost in rewiring or replacing a fuse after operation.
- The current-time characteristics of a fuse cannot always be correlated with that of the protected device.
- On heavy short circuit , discrimination between fuses in series cannot be obtained unless there is considerable differences in the relative sizes of the fuses concerned.

Miniature Circuit Breaker (MCB)

- A miniature circuit breaker (MCB) automatically switches off electrical circuit during an abnormal condition of the network means in overload condition as well as faulty condition.
- Nowadays we use more commonly miniature circuit breaker or MCB in low voltage electrical network instead of fuse.
- Handling an MCB is electrically safer than a fuse.
- The trip unit is the main part, responsible for its proper working.
- There are two main types of trip mechanism.
- A bi-metal provides protection against **overload current** and an electromagnet provides protection against short-circuit current.
- Arc chute consists of a set parallel and mutually insulated metal plates that can extinguish an electric arc. Arc chutes splits, widens, cools and extinguishes the arc.







WHAT IS MCB ?

- MCB is Miniature Circuit Breaker .
- It is commonly used in low voltage electrical network instead of fuse.
- It automatically switches off the electrical circuit during abnormal condition of the network means in over load condition as well as faulty condition.
- As the switch operating knob comes at its off position during tripping, the faulty zone of the electrical circuit can easily be identified.
- It is mostly costlier.



MCB CONSTRUCTION:

- Miniature circuit breaker construction is very simple, robust and maintenance free.
- A miniature circuit breaker has normally three main constructional parts. These are:

Frame of Miniature Circuit Breaker

The Frame of Miniature Circuit Breaker is a molded case.

This is a rigid, strong, insulated housing in which the other components are mounted.

Operating Mechanism of Miniature Circuit Breaker

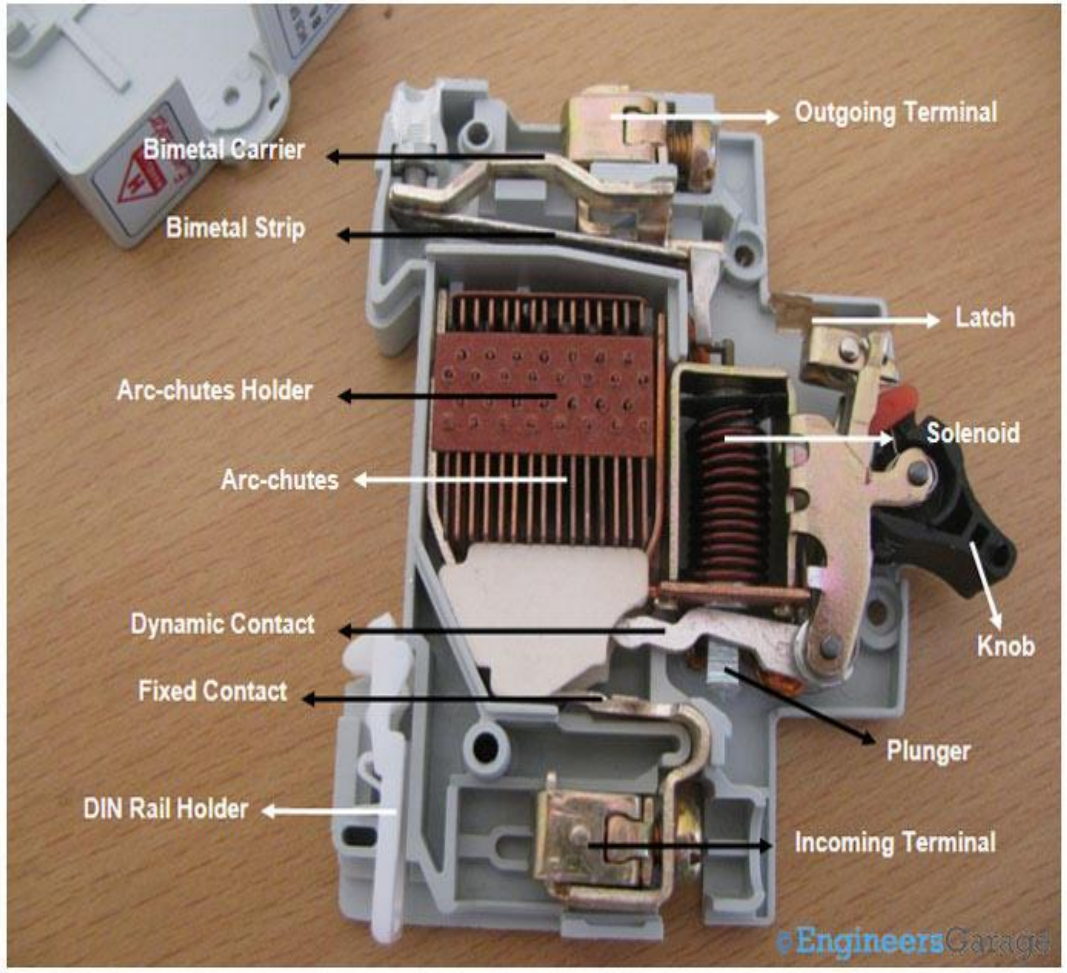
The Operating Mechanism of Miniature Circuit Breaker provides the means of manual opening and closing operation of miniature circuit breaker.

Trip Unit of Miniature Circuit Breaker

The Trip Unit is the main part, responsible for proper working of miniature circuit breaker.

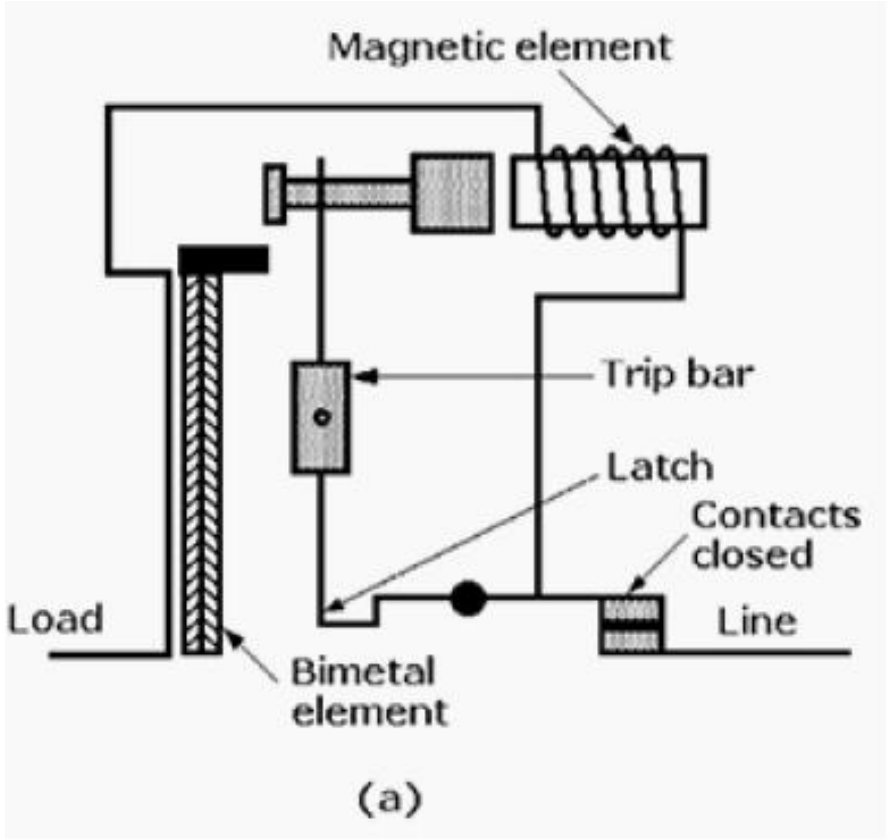
Working of MCB

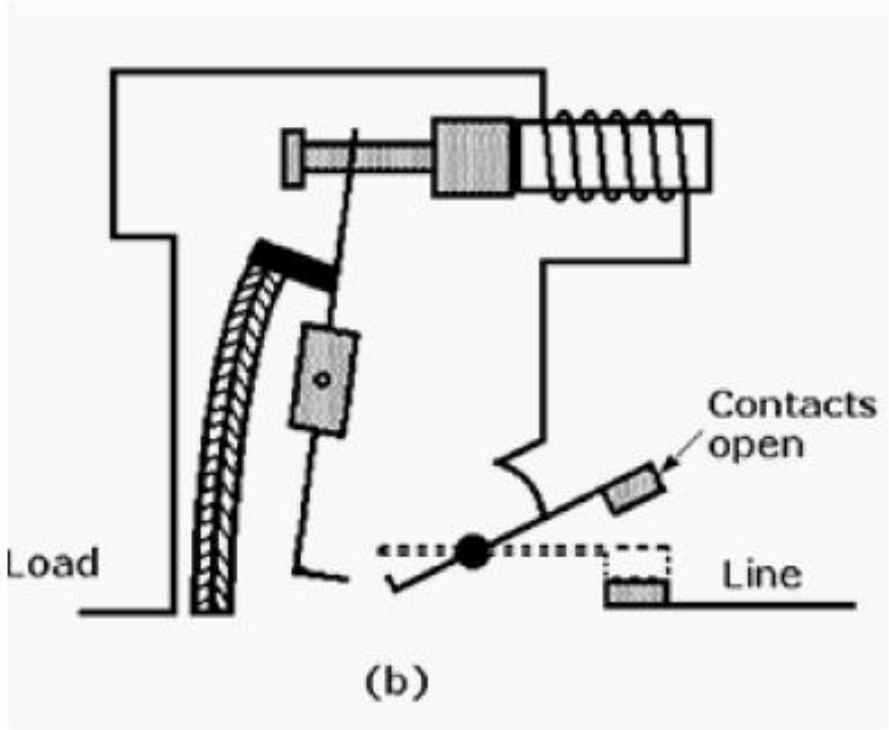
- There are two arrangement of **operation of miniature circuit breaker.**
- One due to thermal effect of over current and other due to electromagnetic effect of over current.
- The thermal **operation of miniature circuit** breaker is achieved with a bimetallic strip whenever continuous over current flows through MCB, the bimetallic strip is heated and deflects by bending.



Working of MCB

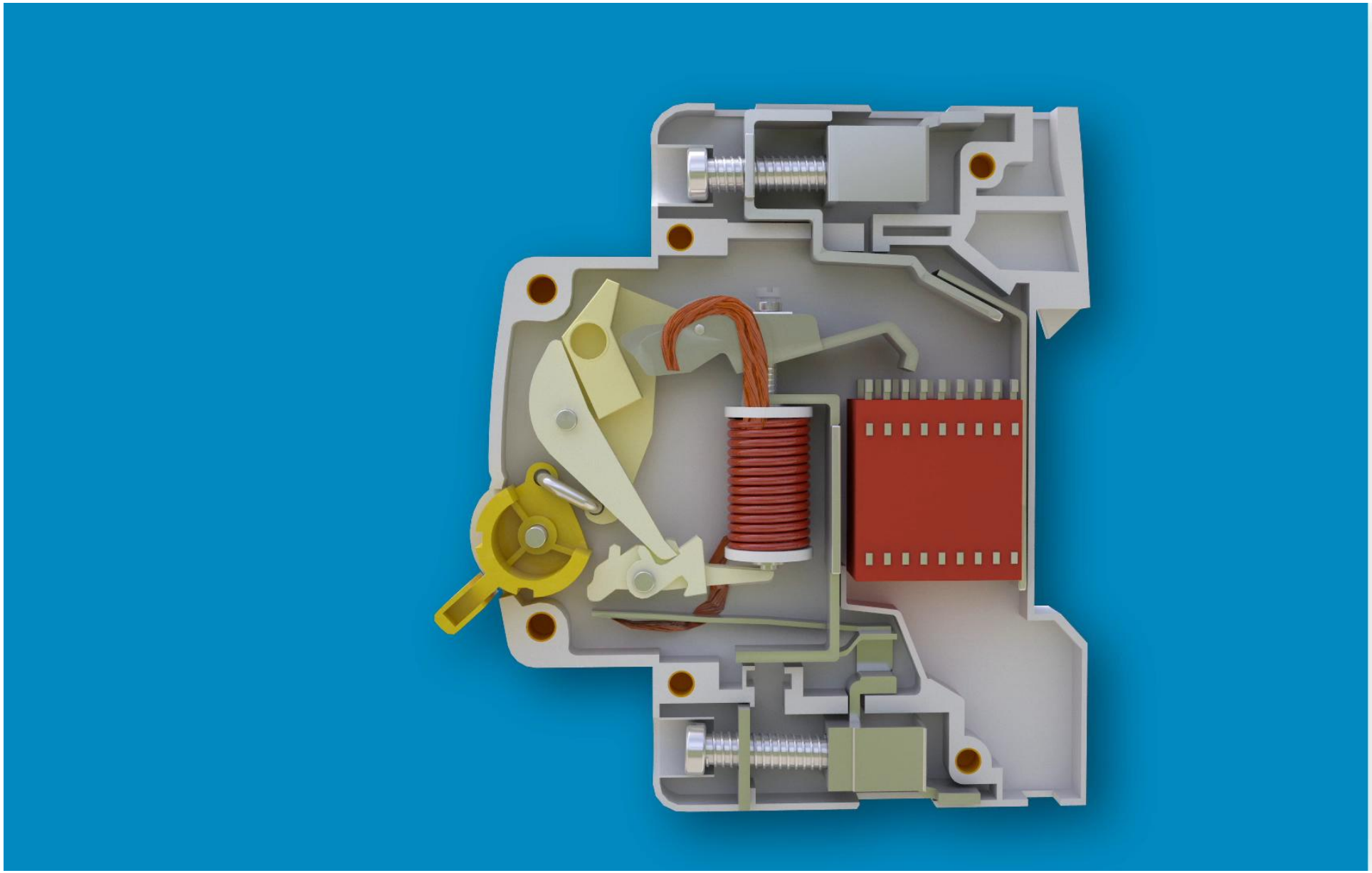
- There are two arrangement of **operation of miniature circuit breaker.**





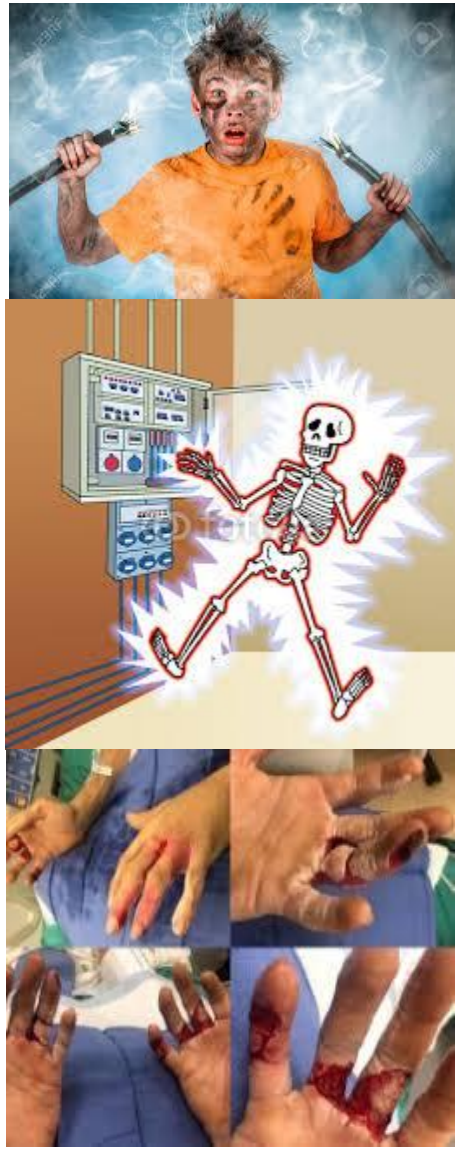
MCB Operation

Miniature Circuit Breaker: Working and Construction (Video)



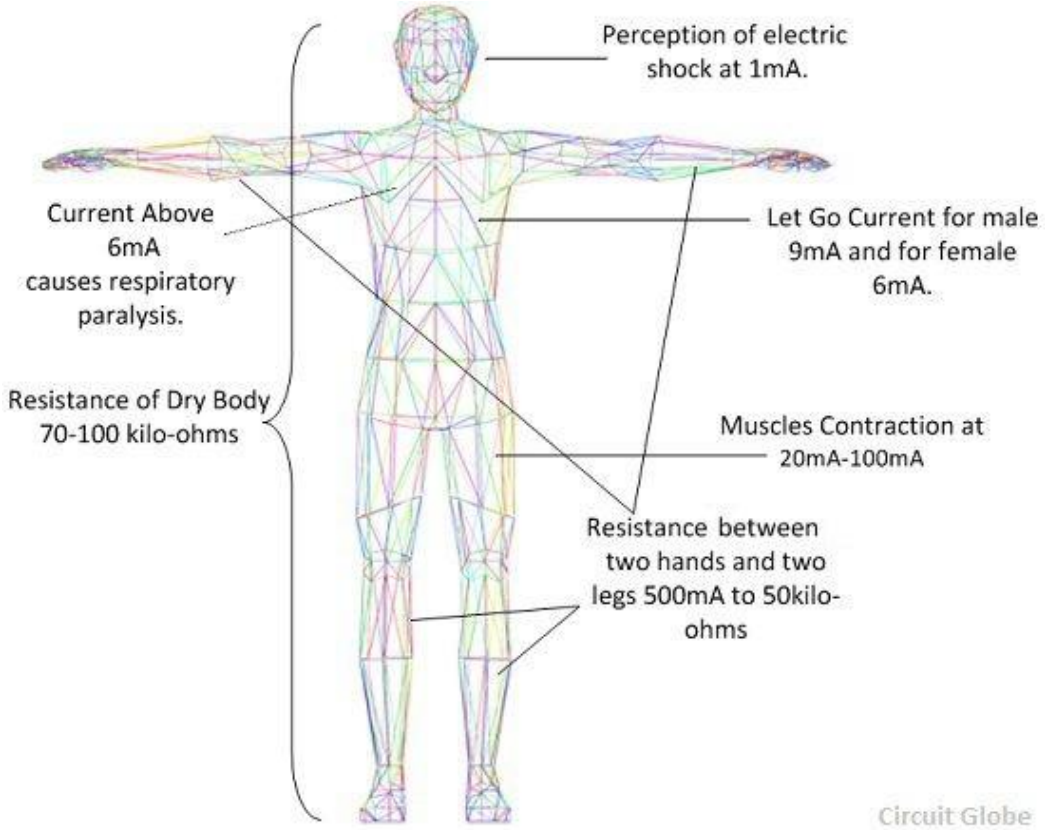
Electric Shock

- Our body is not conductor, it is insulator. Not many but a little bit current can pass through which we can feel and not have harm us. But if large quantity passes through ,our body get a shock.
- An electric shock occurs when a person comes into contact with an electrical energy source.
- Electrical energy flows through a portion of the body causing a shock.
- Exposure to electrical energy may result in no injury at all or may result in devastating damage or death.
- Many variables determine what injuries may occur.
  - These variables include the type of current (AC or DC),
  - The amount of current (determined by the voltage of the source and the resistance of the tissues involved).
  - The pathway the electricity takes through the body.
  - Time of Current passing through the body.
- "Under dry conditions, the resistance offered by the human body may be as high as100,000 Ohms. Wet or broken skin may drop the body's resistance to1,000 Ohms,"



Electric Current on human Body

- The principle that “current kills” is essentially correct.
- It is electric current that burns tissue, freezes muscles, and fibrillates hearts.
- However, electric current doesn't just occur on its own: there must be voltage available to motivate the current to flow through a victim.
- The higher the current, the more likely it is lethal. Since current is proportional to voltage when resistance is fixed (ohm's law),
- high voltage is an indirect risk for producing higher currents. Duration. The longer the duration, the more likely it is lethal



Current in milliamperes	Effects
1 or less	No sensation; probably not noticed
1 to 3	Mild sensation not painful
3 to 10	Painful shock.
10 to 30	Muscular control could be lost or muscle clamping
30 to 75	Respiratory paralysis
75mA to 4 **amps**	Ventricular Fibrillation
Over 4 **amps**	Tissue begins to burns. Heart muscles clamp and heart stops beating

Effects of Electrical Current On the Human Body

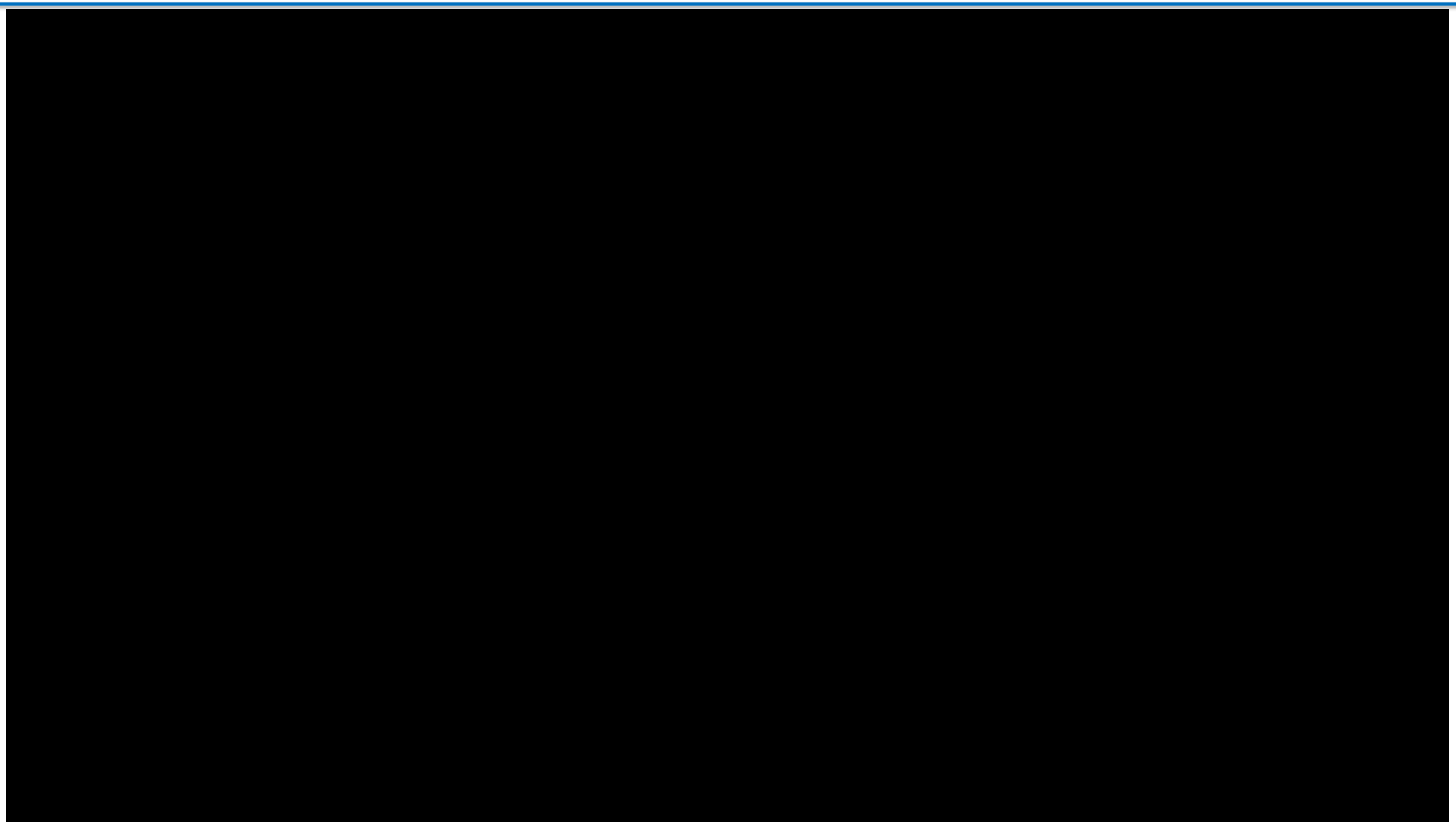


Precautions against Electric Shock

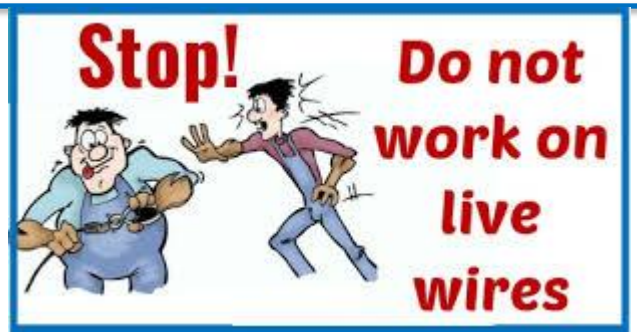
1. Avoid water at all times when working with electricity. Never touch or try repairing any electrical equipment or circuits with wet hands. It increases the conductivity of electric current.
2. Never use equipment with frayed cords, damaged insulation or broken plugs.
3. If you are working on any receptacle at your home then always turn off the mains. It is also a good idea to put up a sign on the service panel so that nobody turns the main switch ON by accident.
4. Always use insulated tools while working.



Electric Shock Safety Animation Video



Safety Precautions When Working With Electricity



1. Never touch or try repairing any electrical equipment or circuits with wet hands. It increases the conductivity of electric current.
2. Never use equipment with damaged insulation or broken plugs.
3. If you are working on any receptacle or electrical Sockets or Outlets at your home then always turn off the mains.
4. Always use insulated tools while working.
5. Always use appropriate insulated rubber gloves and goggles while working on any branch circuit or any other electrical circuit.
6. Never try repairing energized equipment. Always check that it is de-energized first by using a tester.
7. Never use an aluminium or steel ladder if you are working on any receptacle at height in your home. Use a bamboo, wooden or a fibre glass ladder instead.
8. Always use a circuit breaker or fuse with the appropriate current rating.







Earthing : Necessity, Working and Types

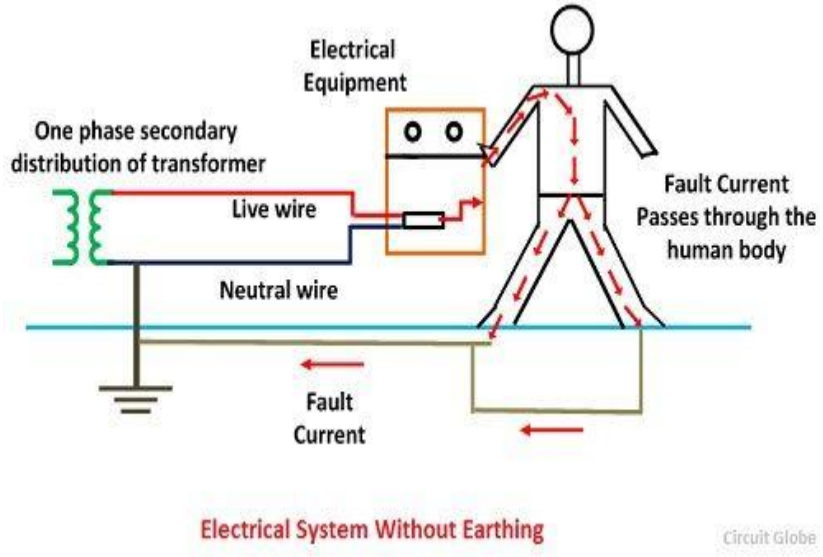
The process of transferring the immediate discharge of the electrical energy directly to the earth by the help of the low resistance wire is known as the electrical earthing.

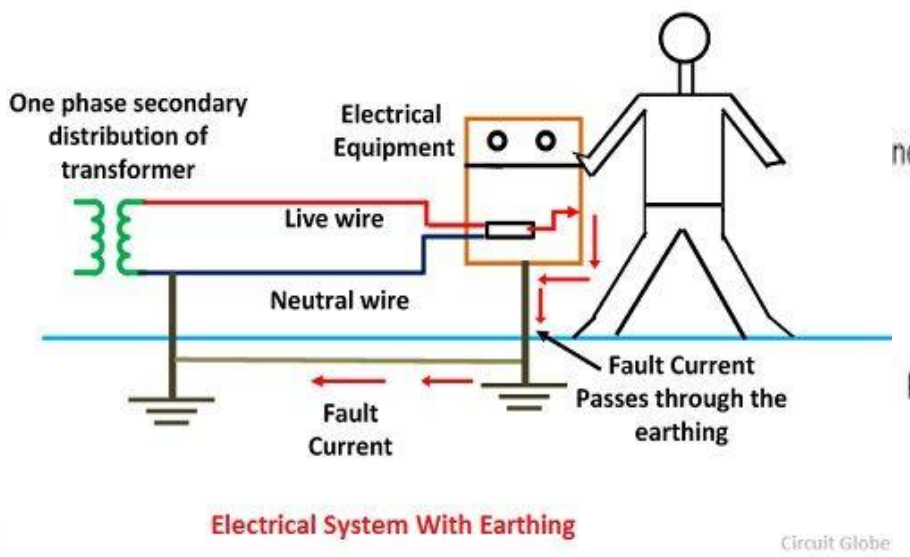


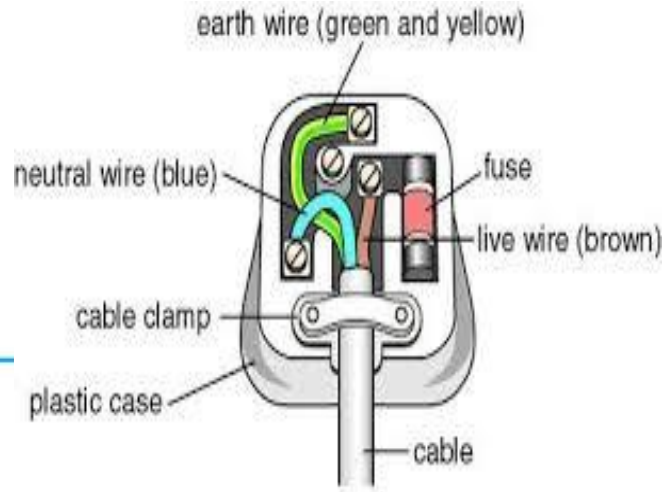
The earthing provides the simple path to the leakage current.

The earthing is essential because of the following reasons

- ✓ The earthing protects the personnel from the short circuit current.
- ✓ The earthing provides the easiest path to the flow of short circuit current even after the failure of the insulation.
- ✓ The earthing protects the apparatus and personnel from the high voltage surges and lightning discharge.







Earthing VS Grounding

Earthing	Grounding
This method protects human beings from electric shocks.	This method protects the entire power system from malfunctioning.
The earth wire used is green in colour.	The wire used for grounding is black in colour
Earthing is primarily used to avoid electric shocks.	Grounding is primarily used for unbalancing when the electric system overloads.
Earthing is located under the earth pit, between the equipment body and the underground pit.	It is located between the neutral of the equipment being used and the ground.

Earthing :

- The method of connecting the body of an equipment to the general mass of earth.
- It brings the body of the equipment to zero potential and thus avoids shocks to the personnel, incase the body of equipment comes in contact with the live wire.
- The neutral of the supply is solidly earthed to ensure that its potential is zero.

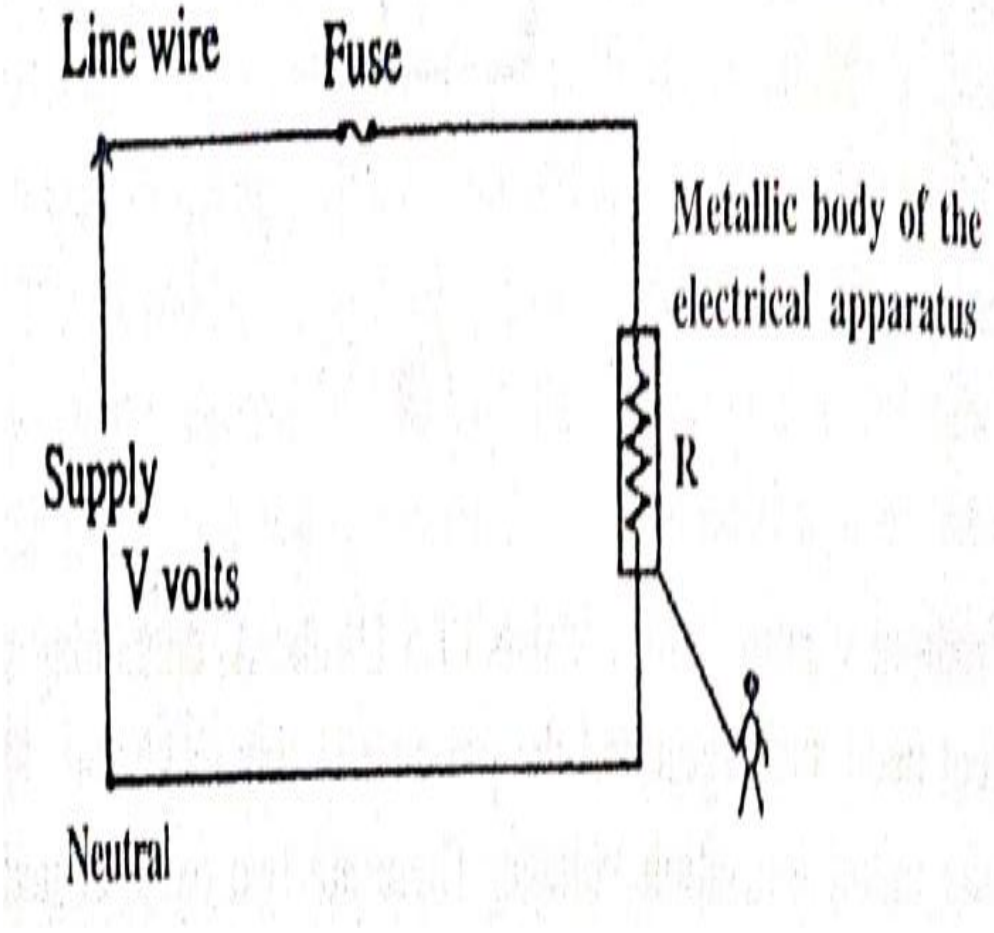


Plate Earthing

- In this method a copper plate of 90cm x 90cm x 0.32cm or a GI plate of the size 90cm x 90cm x 6.35cm is used for earthing.
- The plate is placed vertically down inside the ground at a depth of 3m
- and is embedded in alternate layers of coal and salt for a thickness of 15 cm.
- In addition, water is poured for keeping the earth electrode resistance value well below a maximum of 5 ohms.
- The earth wire is securely bolted to the earth plate.
- A cement masonry chamber is built with a cast iron cover for easy regular maintenance.



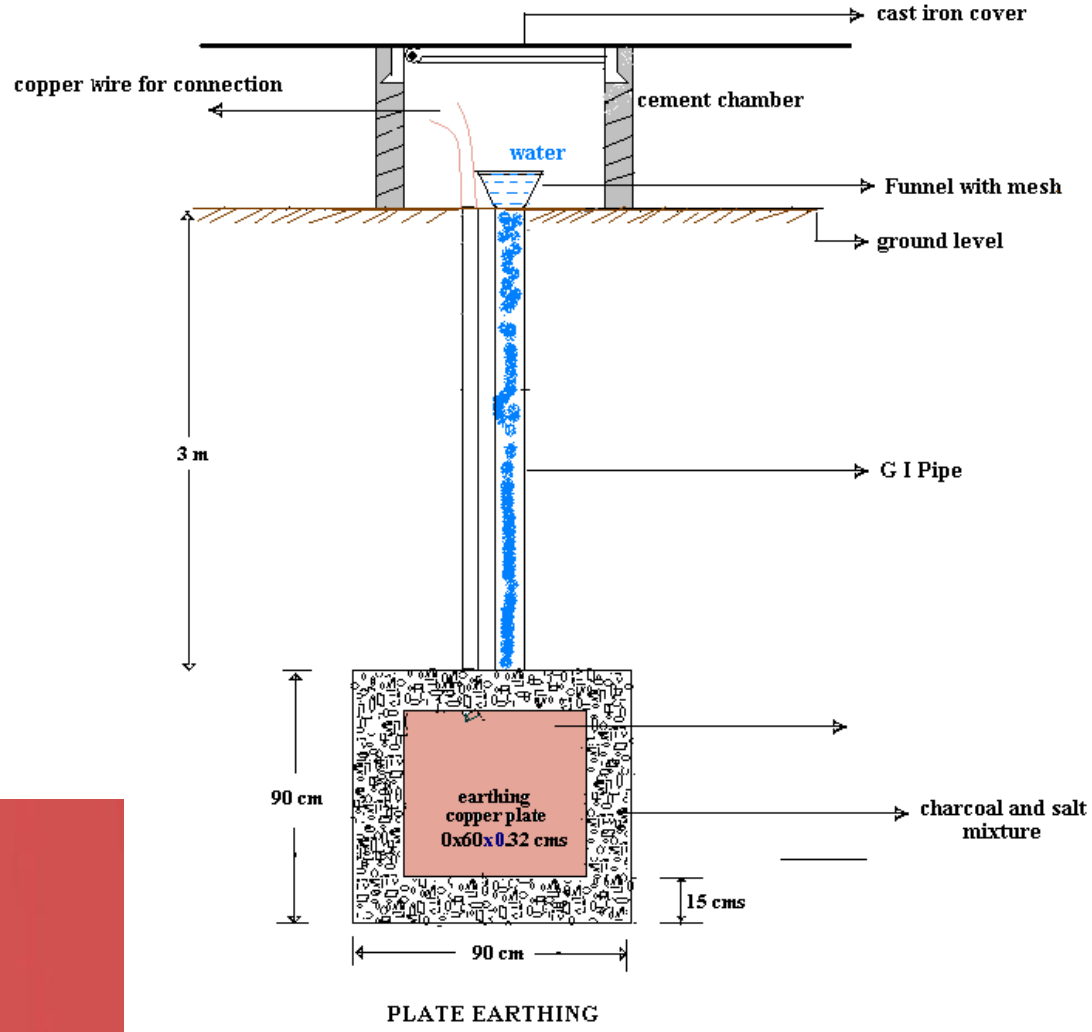
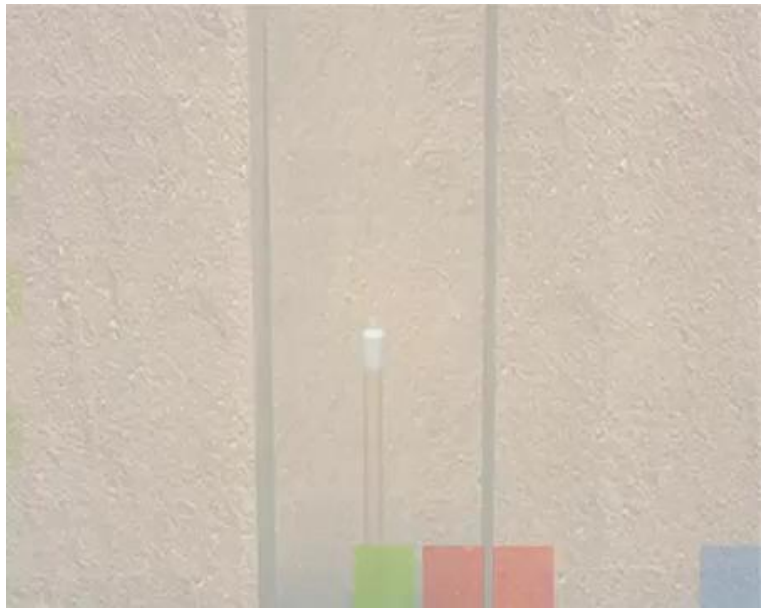


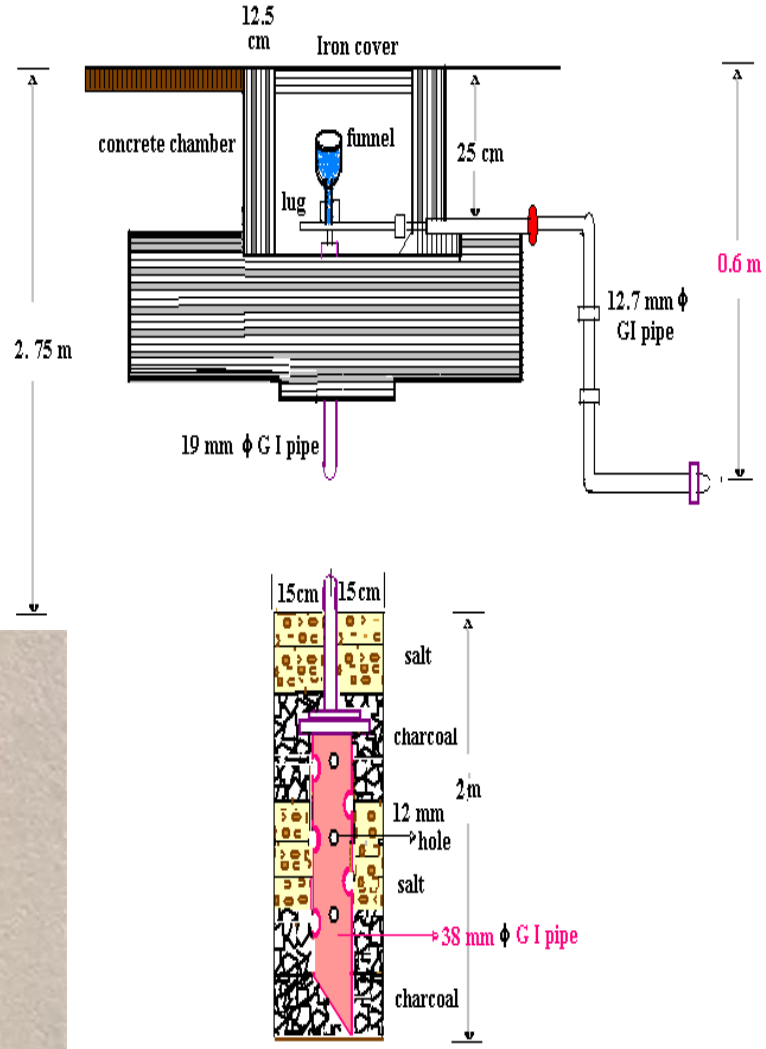
PLATE EARTHING

Pipe Earthing

- Earth electrode made of a GI (galvanized) iron pipe of 38mm in diameter and length of 2m with 12mm holes on the surface is placed upright at a depth of 4.75m in a permanently wet ground.
- To keep the value of the earth resistance at the desired level, the area (15 cms) surrounding the GI pipe is filled with a mixture of salt and coal..
- The efficiency of the earthing system is improved by pouring water through the funnel periodically.

Note: When compared to the plate earth system the pipe earth system can carry larger leakage currents





PIPE EARTHING

Energy Estimation of Common Household Loads.

The Electrical Energy consumed is metered by energy meter connected on the premises of the residence, residential complex or industry. The Energy is measured in terms of Units. One unit of energy is the energy consumed when a load of one kilowatt runs for one hour.

1 Unit = 1 kWhr.

The Domestic consumer is billed for the energy consumed. An industrial consumer is billed both for the load as well for the energy consumed.

Energy Bill Calculator

This will calculate approximate bill for one month	
Supply Type*	LT
Tariff*	Residential

LT Bill Calculator	
Sanction Load (KW)*	5
Connected Load (KW)*	5
Phase*	1 - Phase
Consumption*	169.89

Estimated LT Bill	
Tariff effective from	01 Feb 2014
FAC for the month	Feb 2014

Particulars	**Amount (Rs)**
Fixed Charges (FC)	40.00
Energy Charges (EC)	758.83
Fuel Adjustment Charges (FAC)	0.00
Tax on Sale (TOS) @ 8%	0.00
Electricity Duty (ED) @ 15%	119.83
Total Estimated Bill	**918.66**

Problems

Example 1: Estimate Total Daily Energy Requirement for the following loads.

Name of the Appliance	Power Rating (W)	Avg. Daily Usage Hrs	No. of Appliances
CFL	12	6	3
Fan	50	8	2
TV (21")	150	2	1
Computer	250	3	1

Take electricity cost to be Rs.6 per unit.

Take electricity cost to be Rs.6 per unit.

Solution:

Name of the Appliance	Power Rating (W)	Avg. Daily Usage Hrs	No. of Appliances	Daily Energy Required (Wh)
CFL	12	6	3	216
Fan	50	8	2	800
TV (21")	150	2	1	300
Computer	250	3	1	750
Total Energy				2066 Wh

Hence Monthly Energy Requirement = Daily Energy Required X Days per Month

= 2066 X 30

= **61980 Wh**

= **61.98 kWh or 61.98 Units**

Therefore, the monthly electricity bill is

Monthly Bill = 61.98 units X Rs.6/- per unit

= **Rs.371.88 /-**

2. A geyser is rated at 3kW, 230V, 50Hz. If it is switched ON for one hour daily, what would be the energy cost saving, at the rate of Rs. 2.50 per unit if it is replaced by a solar water heater?

Solution:

Solar water heaters use energy from the Sun. Though their initial investment is high, the running cost is very low and are environmental friendly.

A 3kW geyser running for 1 hour daily would consume 3 units daily. The energy consumed per month is 3 units X 30 days = 90 units per month. The cost of energy per month is 90 units per month X Rs. 2.50 per unit = Rs. 225.

This would be the saving in electricity bill if solar water heater replaces the electric geyser.

3. Estimate the total daily energy requirement for the loads shown in the table below. Also calculate monthly electricity bill if electricity cost is Rs. 5/unit for the first 40 units, rs.4/unit for next 50 units and Rs.2/unit for rest of the units.

Name of the appliance	Power Rating(W)	Avg daily usage hrs.	No of appliances
Fan	50	10	4
CFL	12	6	8
LED TV	100	5	2
Refrigerator	300	24	1
Laptop	60	5	1

Solution:

Name of the appliance	Power Rating(W)	Avg daily usage hrs.	No of appliances	Daily Energy Requiremet (Whr)
Fan	50	10	4	2000
CFL	12	6	8	576
LED TV	100	5	2	1000
Refrigerator	300	24	1	7200
Laptop	60	5	1	300
			Total Energy	11,076 Whr

Hence monthly energy requirement = 11,076 x 30 = 332.28 kWh

= 332.28 units

Monthly electricity bill = 5 x 40 + 4 x 50 + 2 x 242.28 = Rs. 884.56/-

HW…..

4. An electrical installation consists of 13 light points of 80 *W* each, 5 light points of 50 W lamp, 6 fans of 100 *W* capacity and a pump motor of 1 HP. Assuming that 45% of light and fans are used for 6 hours per day and that the water pump works for 2 hours daily. Find out the monthly consumption and cost of electricity bill, based on the tariff of 3 Rs per *KWh*.

5. Estimate the monthly electricity bill for the subsequent load fitted in an electrical installation.

(a) 15 lamps 50 watts each working 4 hours/ day.

(b) 4 ceiling fans 100 watts each working 8 hours/day.

(c) 2 kw heater working 2 hours/day.

(d) Water pump of 0.5 HP runs for 2 hours per day

Rate of charges for light and fans is 3.5 Rs / unit and heater and motor 4 Rs/unit.

6. Calculate the total daily Energy requirement for the following loads. The electricity cost is Rs.6 per unit for the first 20 units, Rs.4 per unit for the next 30 units and Rs.2 per unit for usage above this..

Name of the appliance	Power Rating (W)	Avg. Daily Usage Hours	No. of Appliances
Tube Light	50	6	3
TV	150	3	2
Fan	40	8	2
Computer	300	1	1
AC	1000	4	1

STAR rating of the Appliances



Thank You All